AU OPTRONICS CORPORATION

Specification for Approval

INCOMING INSPECTION STANDARD FOR 14.0" TFT-LCD MODULES (A+ grade)

Model Name: G140HAB01.2

Approved By	Prepared By
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General Display Business Unit/AU Optronics

Customer	Checked and Approved by

Revision History				
Rev.	**Date**	**Page**	**Old Inspection Spec**	**New Inspection Spec**
1	2023.May. 05	All		Preliminary Specification
2	2023.May. 12	9、10	Preliminary Specification	Preliminary Specification add TP
3	2024.July. 09	10		Add TP Pin Hole spec

(Note 1) The revised inspection criteria is effective before the expiration of the IIS as specified

1. Scope:

1.1 The incoming inspection standards shall be applied to TFT-LCD Modules (hereinafter called ""Modules"") that supplied by AUO Display Plus Corporation (hereinafter called ""seller"").

1.2 Specifications contains

- Electrical inspection specification
- Appearance specification
- Outside dimension specification

2. Incoming inspection:

The buyer (customer) shall inspect the modules within twenty calendar days since the delivery date (the ""inspection period"") at its own cost. The results of the inspection (acceptance or rejection) shall be recorded in writing, and a copy of this writing will be promptly sent to the seller.

The buyer may, under commercially reasonable reject procedures, reject an entire lot in the delivery involved. Within the inspection period,if the samples of modules within a lot show a number of unacceptable defects in accordance with this incoming inspection standards, the buyer must notify the seller in writing of any such rejection promptly, and not later than within three business days in the end of the inspection period.

Should the buyer fail to notify the seller within the inspection period, the buyer''s right to reject the modules shall be lapsed and the modules shall be deemed to have been accepted by the buyer.

3. Inspection sampling method:

Unless otherwise agree in writing, the method of incoming inspection shall be based on MIL-STD-105E.

3.1 Lot size: Quantity per shipment lot per model.

3.2 Sampling type: Normal inspection, single sampling.

3.3 Sampling level: Level II.

3.4 Acceptable quality level (AQL):

Major defect: AQL=1.0%.

Minor defect: AQL=2.5%.

Fig.1: Inspection Sampling standard for MIL-STD-105E :

表14-1 MIL-STD-105E 樣本代字

批量	特殊檢驗水準				一般檢驗水準		
	S-1	S-2	S-3	S-4	I	II	III
2~8	A	A	A	A	A	A	B
9~15	A	A	A	A	A	B	C
16~25	A	A	B	B	B	C	D
26~50	A	B	B	C	C	D	E
51~90	B	B	C	C	C	E	F
91~150	B	B	C	D	D	F	G
151~280	B	C	D	E	E	G	H
281~500	B	C	D	E	F	H	J
501~1,200	C	C	E	F	G	J	K
1,201~3,200	C	D	E	G	H	K	L
3,201~10,000	C	D	F	G	J	L	M
10,001~35,000	C	D	F	H	K	M	N
35,001~150,000	D	E	G	J	L	N	P
150,001~500,000	D	E	G	J	M	P	Q
500,000以上	D	E	H	K	N	Q	R

Fig.2 Incoming inspection judgment method :

表14-2 MIL-STD-105E正常檢驗單次抽樣計畫

樣本代字	樣本數	允收品質水準（AQL，%）																									
		0.010	0.015	0.025	0.040	0.065	0.10	0.15	0.25	0.40	0.65	1.0	1.5	2.5	4.0	6.5	10	15	25	40	65	100	150	250	400	650	1000
		Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re	Ac Re
A	2	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	0 1	↓	↓	1 2	2 3	3 4	5 6	7 8	10 11	14 15	21 22	30 31
B	3	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	0 1	↓	↓	1 2	2 3	3 4	5 6	7 8	10 11	14 15	21 22	30 31	44 45
C	5	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	0 1	↑	↑	1 2	2 3	3 4	5 6	7 8	10 11	14 15	21 22	30 31	44 45	↑
D	8	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	0 1	↑	↓	1 2	2 3	3 4	5 6	7 8	10 11	14 15	21 22	30 31	44 45	↑	↑
E	13	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	0 1	↑	↓	1 2	2 3	3 4	5 6	7 8	10 11	14 15	21 22	30 31	44 45	↑	↑	↑
F	20	↓	↓	↓	↓	↓	↓	↓	↓	↓	0 1	↑	↓	1 2	2 3	3 4	5 6	7 8	10 11	14 15	21 22	↑	↑	↑	↑	↑	↑
G	32	↓	↓	↓	↓	↓	↓	↓	↓	0 1	↑	↓	1 2	2 3	3 4	5 6	7 8	10 11	14 15	21 22	↑	↑	↑	↑	↑	↑	↑
H	50	↓	↓	↓	↓	↓	↓	↓	0 1	↑	↓	1 2	2 3	3 4	5 6	7 8	10 11	14 15	21 22	↑	↑	↑	↑	↑	↑	↑	↑
J	80	↓	↓	↓	↓	↓	↓	0 1	↑	↓	1 2	2 3	3 4	5 6	7 8	10 11	14 15	21 22	↑	↑	↑	↑	↑	↑	↑	↑	↑
K	125	↓	↓	↓	↓	↓	0 1	↑	↓	1 2	2 3	3 4	5 6	7 8	10 11	14 15	21 22	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑
L	200	↓	↓	↓	↓	0 1	↑	↓	1 2	2 3	3 4	5 6	7 8	10 11	14 15	21 22	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑
M	315	↓	↓	↓	0 1	↑	↓	1 2	2 3	3 4	5 6	7 8	10 11	14 15	21 22	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑
N	500	↓	↓	0 1	↑	↓	1 2	2 3	3 4	5 6	7 8	10 11	14 15	21 22	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑
P	800	↓	0 1	↑	↓	1 2	2 3	3 4	5 6	7 8	10 11	14 15	21 22	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑
Q	1250	0 1	↑	↓	1 2	2 3	3 4	5 6	7 8	10 11	14 15	21 22	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑
R	2000	↑	↑	1 2	2 3	3 4	5 6	7 8	10 11	14 15	21 22	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑

↓ =採用箭頭下第一個抽樣計畫，如樣本大小等於或超過批量時，則用全數檢驗。
↑ =採用箭頭上第一個抽樣計畫。Ac=允收數，Re=拒收數。

4. Inspection instruments:

4.1 Pattern generator: LD-2000 or equivalent model.

4.2 Video board: AU video board or equivalent. The output of the signal should comply with the specification provided by AU.

4.3 Luminance colorimeter: Topcon BM-7 or equivalent model

5. Inspection environment conditions:

5.1 Room temperature : **20 ~ 25** C.

5.2 Humidity: **65±5**% RH.

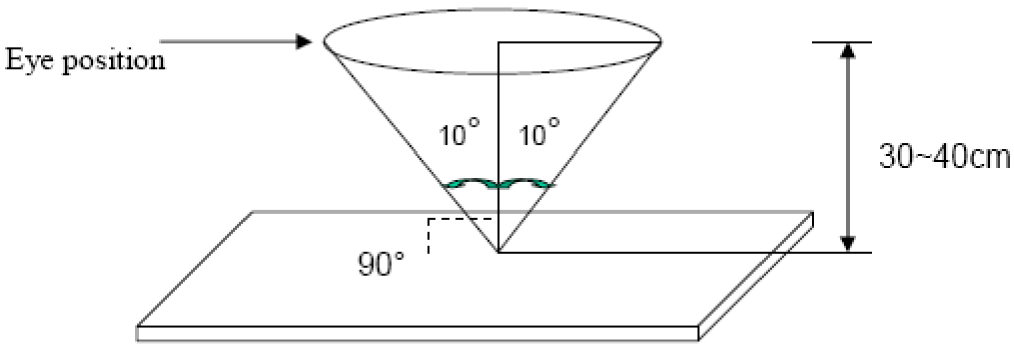
5.3 Illumination: Fluorescent light (Day-Light Type) display surface illumination to be **300 ~ 700** lux. (standard **500** lux.)

5.4 To be a distance about **35 ± 5** cm in front of LCD unit, viewing line should be perpendicular to the surface of the module judge the visual appearance with human's eyes.

5.5 Take off the protector of polarizer while judging the display area.

5.6 If there is any question while judging, check the panel again while operating.

5.7 The judgement by turning on the screen.



6. Classification of defects:

Defects are classified as major defects and minor defects according to the degree of defectiveness defined herein.

Major defects:

A major defect is a defect that is likely to result in failure, or to reduce materially the usability of the product for its intended purpose.

Minor defects:

A minor defect is either a defect that is not likely to reduce materially the usability of the product for its intended purpose, or a departure from an intended purpose with little bearing on the effective use or operation of the product.

6.1 Electrical inspection specification:

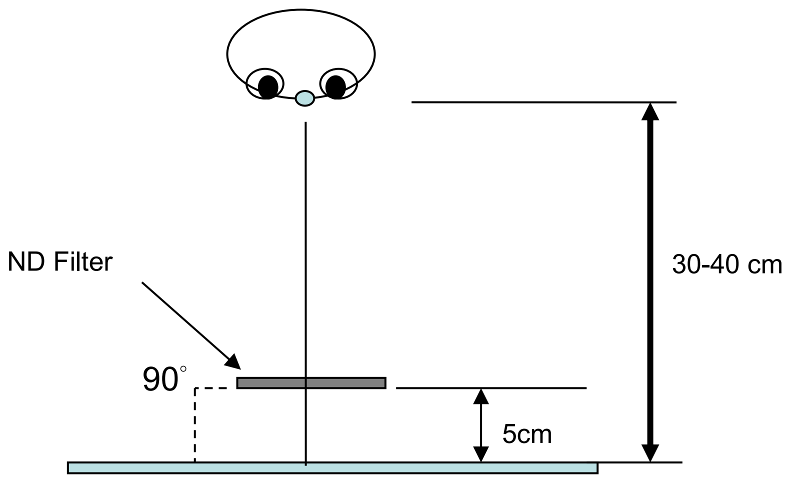
Inspection Item		Specification
Line defect		Not allowed
Bright dots		0 dot
Dark dots		≦4 dots
Total dots defect		≦4 dots
Continuous defect	Two continuous bright dots :	0 pair
	Over three continuous bright dots (vertical, horizontal, oblique) :	Not allowed
	Two continuous dark dots (vertical, horizontal, oblique) :	≦2 pair
	Over three continuous dark dots (vertical, horizontal, oblique) :	Not allowed

	Two continuous dark dots and bright dots (vertical, horizontal, oblique) :	0 pair
	Over three continuous dots (vertical, horizontal, oblique) :	Not allowed
	Distance between 2 Bright dots :	NA
	Distance between 2 Dark dots :	≥ 5 mm
	Distance between Dark dot and Bright Dot :	≥ 5 mm
Mura		5 % ND filter

Note 1) For dot defect, one sub pixel is defined as one dot. Defect area (of dot defect) should be larger than 1/2 area of one sub-pixel to be count as 1 dot defect.

Note 2) Judgment criteria (For Bright dot and Small Bright dot) : Using ND Filter 5% (distance : 30 cm). If it could be observed, dot defines as one bright dot. If not, dot defines as one small bright dot.

ND filter use method: The inspection method of ND Filter - holding ND filter in front of the panel around 5 cm and examine the panel from **35±5** cm in the front view for **3** seconds.



Note 3) A dot defect that is smaller than the defined dot defect will be treated as small bright dot.

The drawing of 1/2 area sub-pixel definition: The 1/2 area sub-pixel can be defined as below one or more of specific shapes (Fig.1). The small bright dots is less than or equal to 20 dots.

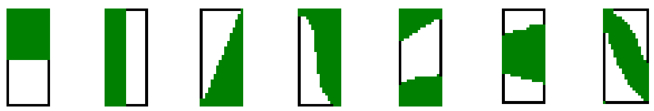


Fig.1

Note 4) All bright dot defects should not be noticeable by observer under specified inspection environment (Please refer to item 5).

Note 5) Adjacent-dot defect should be observed under the same display pattern in any one of Black/Green/Blue/Red pattern.

Definition of two continuous bright dots: Only for two continuous dots (included vertical, horizontal, oblique type)

*Inspection pattern: Standard inspection patterns of dot defect are listed below. AU uses these patterns as standard criteria for judging dot defect. Please inform AU if any other pattern is to be used to examine dot defect.

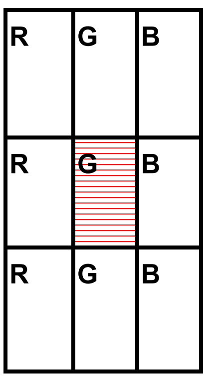
Test Pattern	Defect
Black	For bright dot(s)
Red	For bright and dark dot(s)
Green	For bright and dark dot(s)
Blue	For bright and dark dot(s)

Note 6) In three (or more) adjacent dot defect, for any 3rd dot that adjacent to 2 continuous defective dots (can be of any combination of bright dots and dark dots), the 3rd dot, no matter how large it may be, should be viewed as a dot.

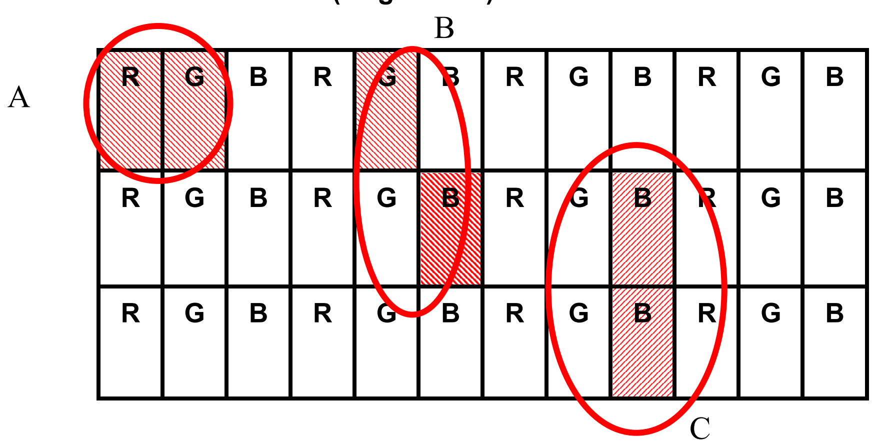
Note 7) Defect criteria diagram

- Dot defect diagram

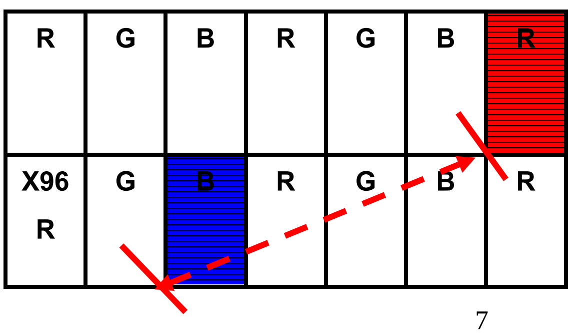
One dot (Bright /Dark)



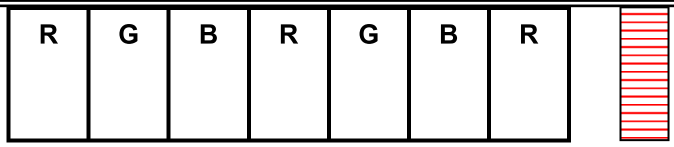
Two continuous dots(Bright/Dark)



- Definition of distance between defect dots as following



A: Distance between defect dots



Defect Dot

Note 8) Unless otherwise specified by written document or limit samples, Mura (display un-uniformity) should inspected under the ND filter and shall be accepted when it is invisible 5 % ND filter is applied.

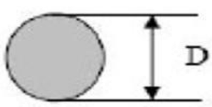
Note 9) While operating over 50℃ ambient temperature, there should be no function failure occur and Mura (display un-uniformity) should be invisible under 1% ND filter applied.

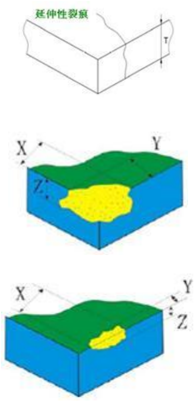
Note 10) Image Retention: Image retention must be disappeared in 1 sec after pattern changed. / Image retention must be disappeared in 3 sec after turn off power.

6.2 Inspection specification

Judge area	Judge item		Inspection specification		Judge criterion: Major	Judge criterion: Minor
Active area	Particles,	Circular	**Average diameter: D (mm)** **Distance: ND(mm)**	**Numbers**		○
			D≦0.15	Disregarded		
			0.15<D≦0.5	n≦4		
			0.5＜D	n=0		
		Linear	**Width: W (mm)** **Length: L (mm)**	**Numbers**		○
			0.05<W≦0.1 0.3≦L≦2	n≦3		
	Scratch/Dent	Circular	**Average diameter: D (mm)**	**Numbers**		○
			D≦0.15	Disregarded		
			0.15＜D≦0.5	n≦4		
			0.5＜D	n=0		
		Linear	**Width: W (mm)** **Length: L (mm)**	**Numbers**		○
			0.05<W≦0.1 0.3≦L≦2	n≦3		
	Bubble	Circular	**Average diameter: D (mm)**	**Numbers**		○
			D≦0.15	Disregarded		
			0.1＜D≦0.5	n≦4		
			0.5＜D	n=0		
Bezel	Gap between front and back bezel on all sides		Gap ≦ 0.5 mm			○
	Scratches, Wrap and Sunken		Allowed (No harm, dangerous)			○
	Assembly Fail		Not allowed		○	
	Color Difference		Allowed (No harm, dangerous)			○
Carton and Panel Label	No label		Not allowed		○	
	Invert label				○	

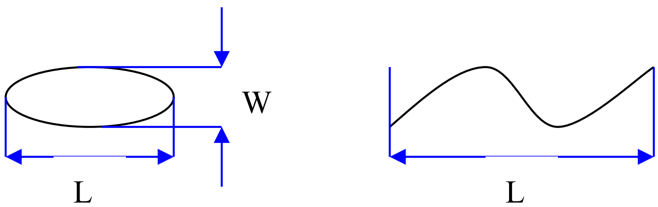
(S/N, B/L, WEEK)	Broken		◯	
	Dirt	No larger than 30mm×20mm.	◯	
	Not clear	Word can be read. Barcode can be scanned.		◯
	Word out of shape			◯
	Content Error	No allowed	◯	
	Position	Be attached on right position		◯
	Crease	Allowed (Word can be read. Barcode can be scanned)	◯	
	Label overlapping	Allowed	◯	
Screw	Not enough (Q'ty)	No allowed	◯	
	Loose	No allowed	◯	
Connector	Appearance	No broken. Deform is allowed if no functional issue.	◯	

Defect item		General Cosmetic Criteria for TP	
		Inspection Criteria	n (quantity)
linear shape defect (scratch/stain/particle) 	VA、BM	W ≦ 0.05mm , L ≦2mm	Ignored(忽略)
		2mm < L < 5mm 0.1mm < W ≦ 0.2mm	N≦5, 間距(Distance) >30mm
		L ≧ 5mm W > 0.2mm NG	NG
dot shape defect (bubble、stain、Newton Ring、black/white spot、dent、particle、scratch) 	VA、BM	D ≦ 0.3mm	Ignored(忽略)
		0.3mm < D≦0.5mm	N≦5, 間距(Distance) >30mm
		D > 0.5mm	NG
printing sawtooth 	VA 邊緣	無基線 S≦0.07mm（相鄰峰谷）不計, 超出 NG； for base line → S≦0.07mm OK；otherwise NG 有基線 S/2≦0.07mm 不計, 超出 NG non-base line →S/2≦0.07mm OK ; otherwise NG	

四邊角落缺角/裂痕/崩邊 Glass flaw / crack 	Corner / Crack	X≦2.0mm Y≦2.0mm Z≦1/2 Glass thickness	N≦2 for each corner
	Edge / Crack	Creeping crack is not allowed. X≦5mm, Y≦0.5mm, Z≦1/2 Glass thickness	N≦2 for each side
凹坑 dent	VA、BM	檢驗視角(45 度)内不可見 the dent invisible within viewing angle is OK 超出檢驗視角,以 Limit Sample 判定 without view angle→ judge by limit sample	
油墨漏光 Pin hole	BM	Dot shape: D≦0.1mm ignored.; 0.1mm＜D≦0.2mm, N≦2, (distance≧10mm) D＞0.2mm, NG Linear Shape: 0.03mm<W≦0.05mm, L≦3mm, N≦2 , (distance≥5mm) 补漆后不可有掉漆/色泽不均匀/或造成透光 It is not allowed to have paint chipping/uneven color/light transmittance after paint touch-up.	

Note 1 : When L>=2W, defect count as liner defect.

Note 2: D=(W+L)/2



Note 3 : Extraneous substances which can be wiped out, such as fingerprint and particles are not considered as a defect.

Note 4 : Defects on the Black Matrix (outside Active Area 0.3mm) are not considered as a defect.

7. Inspection judgement:

7-1 The judgement of the shipped lot (acceptance or rejection) should follow the sampling plan of MIL-STD-105E, single sampling, normal inspection, level II.

7-2 If the number of defects is equal to or less than the applicable acceptance level, the lot shall be accepted.

7-3 If the number of defects is more than the applicable acceptance level, the lot shall be rejected and the buyer should inform the seller of the result of incoming inspection in writing

8 Precaution:

Please pay attention to the following items when you use the LCD Module with back-light unit.

1. During the Module's assembly, do not twist, bend or add unsuitable external force to the Module.
2. Do not use an air gun or blower to clean the Module during the Module's assembly to avoid the contamination of external dust or particles.
3. Only operate the Modules in the specified temperature range and a ventilated environment.
4. Avoid dust or oil mist during Module assembly.
5. Follow the correct power sequence while operating the Module. Module will be damaged otherwise.
6. Avoid electrical magnetic interference for higher safety and lesser noise.
7. Operate the Module in the specified temperature range as the temperature will affect the response time and brightness.
8. Avoid displaying the fixed pattern (especially the white pattern) for a long period of time as it will cause image sticking.
9. Be sure to turn off the power when connecting or disconnecting the circuit.
10. Avoid dirt and stain on the surface of modules. As it can easily scratch the surface polarizer.
11. Wipe off any moistures before operating the Modules as the dewdrop may damage the Modules.
12. Avoid drastic temperature changes, as the condensation or expansion reactions will damage (destroy) the polarizer.
13. Avoid exposing the modules under sunlight as the high temperature and humidity will degrade the performance.
14. Avoid any harmful materials/ components/ chemiclas on the modules, such as acid or chlorine.
15. Connect grounded devices while operating the Module as the static electricity will damage the Module.
16. Disassemble or reassemble the Modules are forbidden.
17. Avoid direct contact on the rear side as the backlight's high voltage will cause electrical shocks.
18. Avoid strong vibration or shock on the Module as they will damage the Module.
19. Store the Modules in suitable environment with regular packing.
20. Do not press on the surface (either front or rear side) of the Module as it will lead to non-uniformity or other function issues. Be cautious of the broken Module to avoid damages.
21. Softly tear the sticking tape of the protective film to prevent bezel out of shape and non-uniform display.

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